of technologies provides a versatile development environment and helps you ramp up the blockchain programming learning curve quickly. Starting in chapter 6, you'll migrate from this initial environment to a production setup that will allow you to develop end-to-end Dapps and deploy them on public blockchains.

On completion of this chapter, you'll be able to analyze a problem, design a smart contract solution, implement it using Solidity, and deploy it on a test blockchain provided by the Remix IDE. You'll also have learned some best practices for blockchain programming.

2.1 The concept of a smart contract

A smart contract is a piece of code that improves on the basic trust enabled by the Bitcoin blockchain protocol. It adds *programmability* that in turn enables transactions for digital assets besides cryptocurrency. A smart contract addresses the need for application-specific verification and validation for blockchain applications. It opens the trust layer of the blockchain for general-purpose applications. Let's explore the smart contract in detail.

I've chosen to discuss Ethereum's definition of a smart contract because Ethereum is a general mainstream blockchain. Also, it has been used as a reference implementation for many other industry blockchains, such as JPMorgan's Quoram (https://www.goquorum.com) blockchain